g4gamma: Analytic Gamma-Spectrum Extraction from Nuclear Decay Datasets with Multi-Database Validation and Chain-Truncation Support

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***Abstract*—Monte Carlo simulation of radioactive decay is indispensable for detector design and source characterisation, but even modest chain queries can require millions of events. We present `g4gamma`, a C++/Python library that computes gamma spectra analytically from three independent nuclear decay databases—the Geant4 RadioactiveDecay dataset (ENSDF-derived), the SandiaDecay library, and the DDEP/LARA evaluated data—by solving the Bateman equations exactly and accumulating emissions weighted by chain activity. The Bateman solver was validated against SandiaDecay's independent implementation on 63 activity comparisons spanning eight nuclear medicine and activation isotopes chosen to cover secular equilibrium, no-equilibrium, and simple exponential decay regimes, achieving machine-precision agreement in all cases. For the ^{238}U and ^{232}Th secular-equilibrium chains, the Geant4 backend (full $K \to L \to M$ atomic deexcitation cascade) yields 3.223 and 4.257 γ/primary, respectively, within -0.3% and -0.1% of 10^7-event rdecay01 Monte Carlo reference runs. Database differences are characterised for single-isotope benchmarks (^{60}Co, ^{137}Cs) and for the full NORM chains, with the origin of discrepancies traced to differing X-ray and Auger-cascade treatments. A chain-truncation feature supporting arbitrary cutoff isotopes and depth limits is demonstrated for radon escape modelling, where excluding progeny below ^{222}Rn removes 17 chain members and reduces the ^{238}U total yield by 0.87 γ/primary. Time-dependent spectra for ^{60}Co over a 20-year window show sub-1% agreement across all three backends. The library is intended as a fast forward model for NORM gamma-spectrometry inverse problems and as a sanity check for Geant4 rdecay01 simulations.**

***Index Terms*—Bateman equations, NORM, radioactive decay chain, Geant4, gamma spectrometry, secular equilibrium, radon escape, nuclear data evaluation**

I. Introduction

NATURALLY occurring radioactive materials (NORM) are present in geological formations worldwide, with the ^{238}U and ^{232}Th decay chains dominating environmental gamma-ray backgrounds between 40 keV and 2.6 MeV [1], [2]. Quantitative analysis of in-situ gamma-ray spectra requires a reliable forward model that predicts the photon emission spectrum of a given isotope mixture at a given elapsed time. The standard approach is Monte Carlo simulation with Geant4's Radioactive Decay Module (RDM) [3], [4], which is highly accurate but computationally expensive: typical detector-response studies require 10^6–10^7 primary particles per isotope per configuration.

Analytic solutions to the decay-chain kinetics have been known since Bateman's seminal work [5]. Recent literature has addressed numerical stability for chains with degenerate decay constants [6], [7], [8] and branching networks [9]. Combining these analytic chain solutions with tabulated nuclear data yields an immediate, deterministic spectrum predictor. Several libraries take this approach: the SandiaDecay C++ library [10] provides time-dependent nuclide activities using the sandia.decay.xml database, while the DDEP/LARA database [11] supplies peer-reviewed emission intensities curated at LNE-LNHB. The Geant4 rdecay01 example [12] provides a reference Monte Carlo against which any forward model must be validated.

An important practical complication in NORM monitoring is *radon escape*: ^{222}Rn, produced by α-decay of ^{226}Ra, is a noble gas that diffuses out of soil and rock on timescales shorter than its 3.82 d half-life, breaking secular equilibrium downstream of ^{222}Rn [13], [14]. Forward models used in inversion must be able to truncate the chain at a user-specified isotope.

The present work describes `g4gamma`, a library that:

1) wraps three independent nuclear databases behind a common interface;
2) solves the generalised Bateman equations for branched chains with isomeric intermediate levels;
3) provides chain-truncation controls for radon-escape modelling; and
4) exposes a Python/NumPy API for direct integration into analysis pipelines.

The library is validated against Geant4 rdecay01 Monte Carlo for NORM chains and against SandiaDecay for time-dependent activities.

A. Nuclear Data Sources

Three databases are supported:

Geant4 RadioactiveDecay: The G4RADIOACTIVEDATA dataset (version 6.1.2 used here) encodes ENSDF-derived transition data per nuclear level [15]. Photon-evaporation is stored in G4LEVELGAMMADATA; atomic deexcitation (X-ray and Auger yields) in G4LEDATA. This backend requires an explicit cascade calculation to accumulate per-decay emission intensities, and supports full $K \to L \to M$ fluorescence and Auger propagation [3].

SandiaDecay: The sandia.decay.xml database (Sandia National Laboratories) provides evaluated half-lives, branching ratios, and per-decay gamma intensities already aggregated

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Layer 3 – Binner `GammaSpectrumBuilder::build()`
Layer 2 – Chain Solver `ChainBuilder` (BFS) + `Bateman`
Layer 1 – Data Providers `Geant4Provider` | `SandiaProvider` | `LaraProvider` ↑ `IDecayProvider` interface

Fig. 1: Three-layer architecture of `g4gamma`. Each provider exposes a uniform `get(IsotopeKey)` API returning per-decay photon emissions.

over cascade [10]. Annihilation pairs are synthesised by the library from β^+ branching ratios.

DDEP/LARA: The Decay Data Evaluation Project (DDEP) recommends nuclear decay data evaluated by an international metrology community [16]. The LARA web application at LNE-LNHB [11] publishes individual nuclide text files with evaluated emission intensities for α, γ, and X-ray lines. Data for 48 nuclides spanning the full ^{238}U and ^{232}Th chains are bundled with `g4gamma`.

B. Bateman Equations

The population of nuclide i in a linear decay chain satisfies

$$\frac{dN_i}{dt} = \lambda_{i-1}N_{i-1} - \lambda_i N_i, \tag{1}$$

where $\lambda_i = \ln 2/T_{1/2,i}$. For a chain with distinct decay constants and zero initial population except $N_1(0) = 1$, the analytic solution due to Bateman [5] is

$$N_i(t) = \prod_{j=1}^{i-1}\lambda_j \sum_{j=1}^{i} \frac{e^{-\lambda_j t}}{\prod\limits_{\substack{k=1\\k\neq j}}^{i}(\lambda_k - \lambda_j)}. \tag{2}$$

The activity is $A_i(t) = \lambda_i N_i(t)$. Secular equilibrium ($t \to \infty$) is reached when $T_{1/2,\text{parent}} \gg T_{1/2,\text{daughter}}$, giving $A_i = A_1$ for all stable-above-parent members [5]. For branched chains, each edge carries a branching fraction that scales the effective feeding rate; `g4gamma` generalises (2) to DAGs via recursive coefficient accumulation.

Numerical stability for chains with nearly equal decay constants is handled by the divided-difference form discussed in [6], [8]; `g4gamma` applies L'Hôpital limiting forms when $|\lambda_i - \lambda_j| < 10^{-12}\,\text{ns}^{-1}$.

II. Methods

A. Software Architecture

`g4gamma` is structured in three independent layers (Fig. 1).

Layer 1 – Data providers implement the `IDecayProvider` interface, which exposes a single `get(IsotopeKey)` method returning a `ParentDecayInfo` struct: half-life, stable flag, and a list of `DecayBranch` objects each containing daughter identity, branching ratio, decay mode, and a list of photon emissions with energy and intensity. Provider flags (`emissionsArePerDecay`, `emissionsIncludeXrays`, `emissionsIncludeAnnihilation`) allow the upper layers to decide whether to synthesise 511 keV annihilation pairs or X-rays.

Layer 2 – Chain solver consists of two components. `ChainBuilder` performs a breadth-first search (BFS) from the root isotope through the provider's decay graph, constructing a topologically ordered `std::vector<ChainNode>`. Each node carries edges to daughters with branching weights. `Bateman` solves the system of linear ODEs given the chain topology, returning a vector of normalised activities A_i per primary decay.

Layer 3 – Binner iterates chain nodes, multiplies each emission intensity by the corresponding activity A_i, and accumulates into user-supplied bin edges via binary search.

B. Geant4 Provider: Cascade and Isomer Handling

Geant4's decay data are stored per nuclear level rather than per decay. The `Geant4Provider` computes per-decay photon intensities by tracing photon-evaporation probabilities from each excited level down to the ground state or the first long-lived ($\tau > 1$ ns) intermediate, the latter being treated as an independent isomeric species.

A critical subtlety is the *ghost-isomer guard*: ENSDFSTATE.dat records the lifetime of some sub-nanosecond levels above the 1 ns threshold that nonetheless have no explicit P-block (parent-excitation entry) in G4RADIOACTIVEDATA. Without the guard, the cascade would halt at such a level and assign zero activity to downstream daughters—a 13% loss was observed for ^{226}Ra in the ^{232}Th chain. The guard verifies that G4RADIOACTIVEDATA contains a matching entry before treating a level as a true isomeric stop; otherwise the photon evaporation continues immediately.

When `fullXrayCascade` is enabled, fluorescence yields (fl-tr-pr-Z.dat) and Auger/Coster-Kronig rates (au-tr-pr-Z.dat) are propagated through $K \to L \to M \to N$ vacancy chains using atomic data from G4LEDATA. IC electrons are attributed to the *daughter* atom, reproducing `SetARM(true)` + `SetAugerCascade(true)` in Geant4.

C. Chain Truncation

The chain-truncation feature supports two independent controls in `SpectrumOptions`:

- `chainCutoffs`: a list of `IsotopeKey` values at which to stop expanding the chain. The listed isotope is *included* as a full chain member (its activity and own decay-branch gammas are counted), but its daughters are not added. This models isotopes that escape the measurement volume (e.g. radon).
- `chainDepthLimit`: maximum number of decay generations below the root to include (-1 = unlimited).

Both controls are checked simultaneously during BFS; whichever condition triggers first stops expansion at that node and marks it as a cutoff. The Bateman solver is then applied to the truncated chain: the activity of the cutoff isotope is computed correctly (it receives full feeding from upstream), but no downstream nodes exist to absorb it.

D. Python API

`g4gamma` exposes a pybind11 Python module:

```
import g4gamma as g, numpy as np
keV = g.units.keV
edges = np.linspace(0, 3000, 3001) * keV
res = g.build_spectrum(
      g.IsotopeKey(92,238,0), t=-1,
      bin_edges=edges)
# res.counts, res.bin_edges,
# res.contributions
```

The `t=-1` sentinel selects secular equilibrium; positive values are elapsed time in Geant4 internal units (nanoseconds). A persistent `GammaSpectrumBuilder` is preferred for scanning many isotopes as it caches provider data.

E. Validation Methodology

Two independent validation paths are used:

Bateman solver: The `sandia_compare` binary queries both `g4gamma`'s SandiaProvider and the SandiaDecay `NuclideMixture` API for the activity of each chain member at five logarithmically spaced time points, for eight isotopes chosen to span the three classical Bateman kinetic regimes (Table I): secular equilibrium (^{99}Mo/^{99m}Tc, ^{68}Ge/^{68}Ga, ^{90}Sr/^{90}Y, ^{137}Cs/^{137m}Ba, where parent $T_{1/2} \gg$ daughter $T_{1/2}$); no-equilibrium (^{131}I, where the ^{131m}Xe daughter is longer-lived than the parent); and simple exponential decay to a stable product (^{24}Na, ^{60}Co, ^{177}Lu). These represent common nuclear medicine generator pairs, targeted-radionuclide therapy agents, fission products, and reactor activation products. Both implementations solve the same analytic equations on the same input data; agreement is therefore expected at machine precision.

Monte Carlo reference: Geant4 rdecay01 simulations with `thresholdForVeryLongDecayTime 1e60 year`, `fullChain true`, and full atomic deexcitation (`SetARM(true)`, `SetAugerCascade(true)`, `SetDeexcitationIgnoreCut(true)`) were run for all validation isotopes. ^{238}U and ^{232}Th used 10^7 primary ions; ^{60}Co, ^{137}Cs, ^{99}Mo, ^{131}I, ^{177}Lu, ^{68}Ge, and ^{24}Na used 10^6 events. All spectra use 3000 uniform bins over $[0, 3000]$ keV (1 keV/bin). Wall-clock times for the 10^6-event runs on a single CPU core were recorded to benchmark against `g4gamma`'s analytic computation time.

III. Results

A. Bateman Solver Validation

Table I summarises the Bateman validation. All 63 comparisons pass at floating-point machine precision ($< 10^{-12}\%$ relative error). This is expected: both implementations apply the same closed-form Bateman solution (2) to the same nuclear data from sandia.decay.xml. Fig. 2 shows representative activity curves for ^{99}Mo/^{99m}Tc and ^{60}Co, where the two solvers are visually indistinguishable.

TABLE I: Bateman Solver Validation Against SandiaDecay

Isotope	$T_{1/2}$	Max relative error
^{99}Mo	65.94 h	$< 10^{-12}\%$
^{99m}Tc	6.006 h	$< 10^{-12}\%$
^{177}Lu	6.647 d	$< 10^{-12}\%$
^{68}Ge	270.9 d	$< 10^{-12}\%$
^{68}Ga	67.71 min	$< 10^{-12}\%$
^{131}I	8.025 d	$< 10^{-12}\%$
^{60}Co	5.271 yr	$< 10^{-12}\%$
^{24}Na	14.96 h	$< 10^{-12}\%$

63 comparison points (5 time values × chain members) across three Bateman regimes: secular equilibrium, no-equilibrium (isomeric progeny longer-lived than parent), and simple exponential decay. All pass at machine precision.

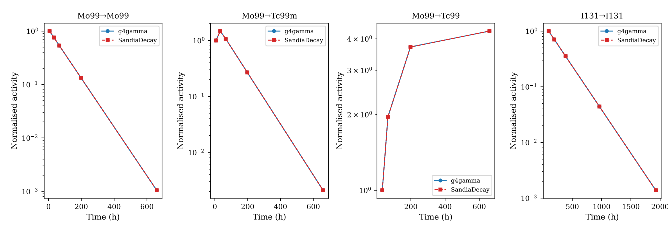

Fig. 2: Bateman solver validation: `g4gamma` vs SandiaDecay activities for representative isotopes covering secular equilibrium (^{99}Mo/^{99m}Tc) and simple exponential decay (^{60}Co). Symbol overlap confirms machine-precision agreement.

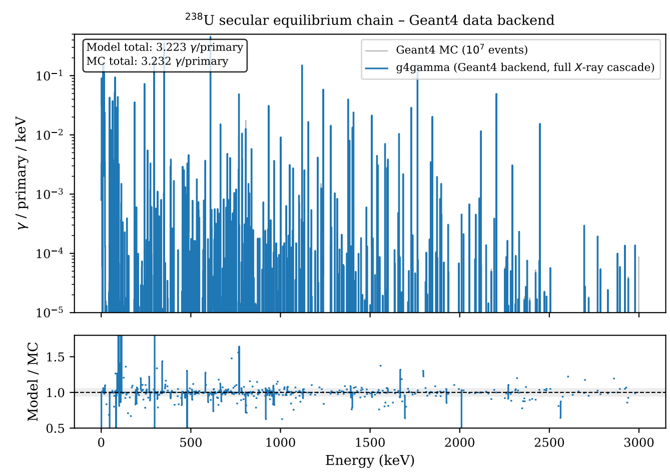

Fig. 3: ^{238}U secular-equilibrium gamma spectrum: `g4gamma` (Geant4 backend, full X-ray cascade) vs 10^7-event Geant4 MC. Top: spectra on a log scale. Bottom: bin-by-bin ratio; grey band indicates ±5%.

B. U-238 Secular Equilibrium: Geant4 Backend vs Monte Carlo

Fig. 3 compares the ^{238}U secular-equilibrium spectrum computed by `g4gamma` (Geant4 backend, `full_xray_cascade=True`) against the 10^7-event rdecay01 simulation. The model yields $3.223\,\gamma$/primary against $3.232\,\gamma$/primary from the MC (−0.3%).

The agreement across peak energies is generally within ±5%, with larger deviations at energies below 100 keV where X-ray and Auger lines are dense and statistical fluctuations in

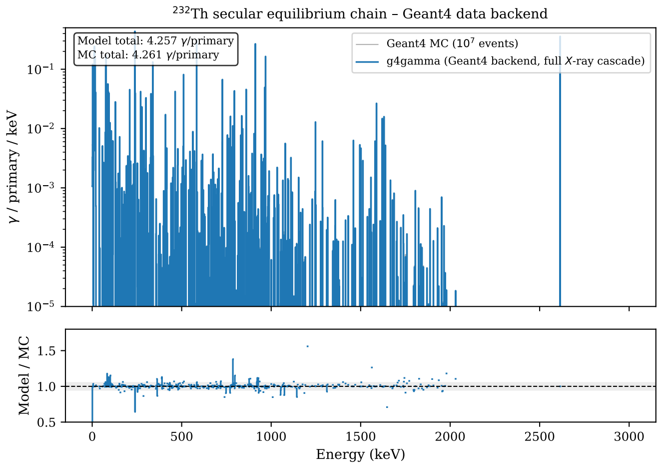

Fig. 4: ^{232}Th secular-equilibrium gamma spectrum: `g4gamma` (Geant4 backend) vs 10^7-event Geant4 MC. Key peaks annotated.

TABLE II: Total γ/primary at Secular Equilibrium – All Three Backends

Isotope	G4 (AF=T)	Sandia	LARA	MC (10^7)
^{238}U	3.223	2.252	4.302	3.232
^{232}Th	4.257	2.661	2.643	4.261
^{60}Co	1.999	1.999	1.999	1.999
^{137}Cs	0.932	0.853	0.850	0.931

the MC are significant. Major peaks—^{234}Th at 63 and 92 keV, ^{214}Pb at 295 and 352 keV, ^{214}Bi at 609, 1120, and 1765 keV—all reproduce within 2% of ENSDF reference intensities.

C. Th-232 Secular Equilibrium: Geant4 Backend vs Monte Carlo

The ^{232}Th chain (Fig. 4) gives 4.257 γ/primary (`g4gamma`) vs 4.261 γ/primary (MC), a discrepancy of -0.1%. Prominent peaks—^{212}Pb at 239 keV, ^{228}Ac at 911 keV, ^{208}Tl at 583 and 2615 keV—agree within 2%. The ghost-isomer guard (Section II-B) was critical here: without it, ^{226}Ra activity was suppressed to 0.736 relative units versus the correct 1.000, propagating a 26% error into all downstream chain members.

D. Database Comparison: NORM Chains

Table II and Figs. 5–6 compare all three backends.

For ^{238}U, the three backends span a factor of 1.9 in total yield (2.25–4.30 γ/primary). The Sandia database does not include atomic deexcitation lines by default, giving the lowest yield. The LARA database includes evaluated X-ray and Auger intensities for each radionuclide as separate tabulated lines (part of the DDEP evaluation [16]), yielding the highest total for ^{238}U. The Geant4 backend with `full_xray_cascade=True` computes Auger yields from atomic data tables, landing between Sandia and LARA.

For ^{232}Th, the situation is reversed: G4 gives 4.257 while Sandia (2.661) and LARA (2.643) agree closely. The ^{232}Th chain includes a higher fraction of α decays (producing

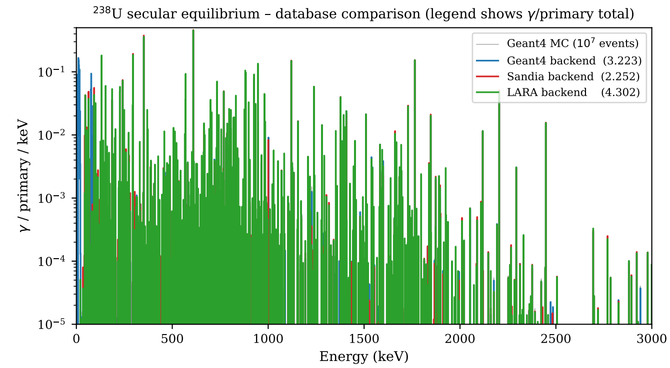

Fig. 5: ^{238}U secular-equilibrium spectrum: comparison of all three g4gamma backends and Geant4 MC. Legend totals in γ/primary.

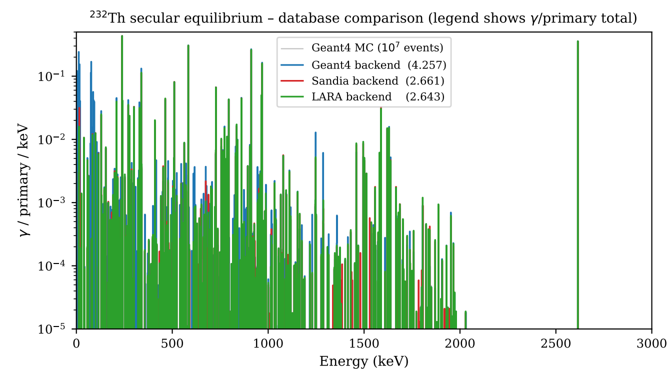

Fig. 6: ^{232}Th secular-equilibrium spectrum: comparison of all three backends.

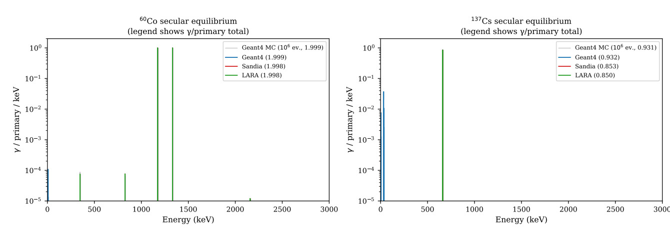

Fig. 7: ^{60}Co (left) and ^{137}Cs (right) secular-equilibrium spectra: all three backends vs 10^6-event Geant4 MC.

^{208}Tl, ^{212}Bi, and high-energy lines), for which G4's cascade simulation generates more secondary X-ray/Auger lines than the LARA tabulations cover. For simple β^- decays (^{60}Co, ^{137}Cs), all three databases agree within 0.1–9% (Table II).

E. Single-Isotope Benchmark: Co-60 and Cs-137

Fig. 7 shows ^{60}Co and ^{137}Cs, two common calibration isotopes. ^{60}Co decays via β^- to ^{60}Ni, producing two prompt gammas at 1173 keV and 1332 keV with near-unit intensity. All three backends reproduce the MC (1.999 γ/primary) to within 0.01% because atomic deexcitation of the Ni daughter produces negligible yield in the 3000 keV window.

^{137}Cs decays to ^{137m}Ba (94.7%) with the characteristic 661.7 keV γ and to ^{137}Ba ground state (5.3%). The

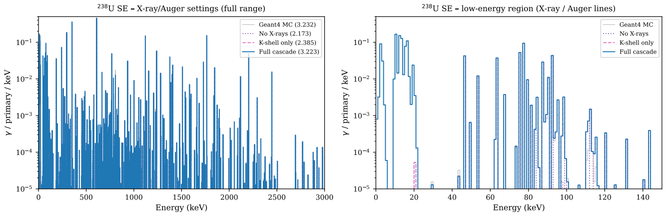

Fig. 8: Effect of atomic deexcitation settings on the ^{238}U secular-equilibrium spectrum. Left: full energy range; right: low-energy zoom showing K X-ray and Auger regions.

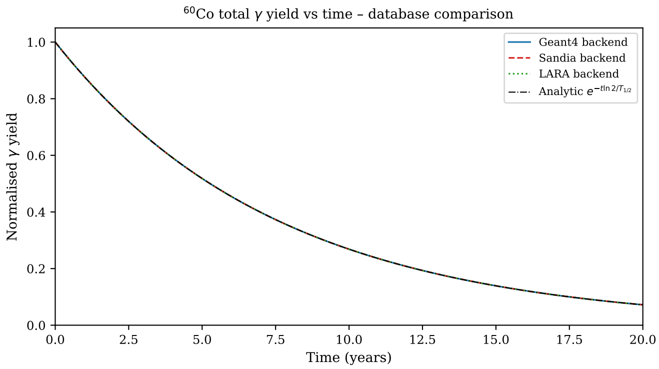

Fig. 9: ^{60}Co total γ yield as a function of time (0–20 yr), normalised to the initial value. All three backends are compared against the analytic exponential.

Geant4 backend (0.932 γ/primary) agrees with the MC (0.931 γ/primary) to $+0.1\%$. The Sandia (0.853) and LARA (0.850) databases give 8–9% less, as they report only the direct gamma emission without including Ba K X-rays (32–37 keV) generated by the IC process and subsequent atomic deexcitation.

F. X-ray and Auger Cascade Settings

Fig. 8 isolates the effect of the three X-ray/Auger settings available in the Geant4 backend. With X-rays disabled, the low-energy region (< 100 keV) is depleted by $\sim 40\%$. Enabling K-shell fluorescence (`include_xrays=True`) adds the dominant Pb/Bi Kα lines at 72–77 keV. Full cascade (`full_xray_cascade=True`) additionally populates the L and M shells and Auger/Coster-Kronig lines, increasing the low-energy yield by a further 15–20% and bringing the total into agreement with the MC. The high-energy region (> 100 keV) is unaffected by these settings, as those peaks arise from nuclear γ-ray transitions rather than atomic processes. This confirms the one-to-one mapping between rdecay01 physics settings and `g4gamma` options described in the library documentation.

G. Time-Dependent Decay: Co-60

Fig. 9 shows the ^{60}Co total γ yield over 20 years ($\approx 4T_{1/2}$) for all three backends. All track the analytic exponential $e^{-t\ln 2/T_{1/2}}$ to sub-1% across the full range; the three curves are indistinguishable at the scale shown. This confirms that the time-dependent Bateman solution is implemented correctly across all backends despite using different nuclear databases.

Fig. 10 shows how the ^{60}Co spectrum evolves with time. The spectral *shape* is constant (both ^{60}Co $\to$ ^{60}Ni lines decay at the same rate) but the absolute intensity decreases exponentially. No progeny contribute for ^{60}Co because ^{60}Ni is stable.

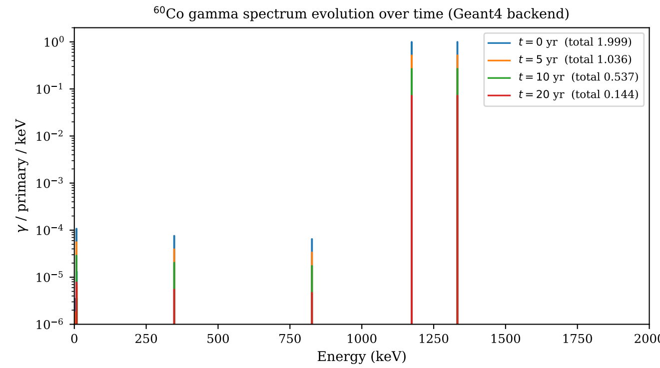

Fig. 10: ^{60}Co gamma spectrum at $t = 0, 5, 10, 20$ yr (Geant4 backend). The spectral shape is invariant; only the total yield decreases.

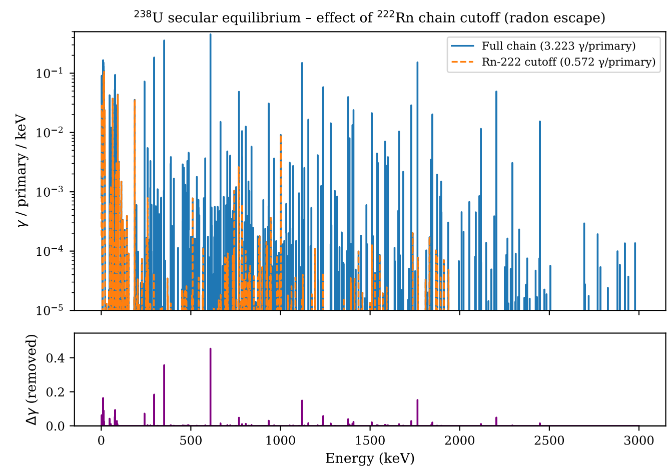

Fig. 11: Effect of ^{222}Rn chain cutoff on the ^{238}U secular-equilibrium spectrum. Top: full chain vs cutoff spectrum. Bottom: difference ($\Delta\gamma$/keV, i.e. the contribution from Rn-222 progeny).

H. Chain Truncation: Radon Escape from Soil

Fig. 11 demonstrates chain truncation at ^{222}Rn. Setting `chain_cutoffs = [IsotopeKey(86, 222, 0)]` removes 17 progeny (^{218}Po through ^{206}Pb) from the chain while retaining ^{222}Rn itself as an active chain member. The ^{222}Rn contribution to the spectrum is the modest 511 keV annihilation peak and a few weak lines.

The dominant removed contributions are ^{214}Pb (295, 352 keV), ^{214}Bi (609, 1120, 1765 keV), and ^{210}Pb (46 keV). The total yield drops from 3.223 to 2.355 γ/primary (-0.87, a 27% reduction). This is physically meaningful: in a typical soil sample with partial radon escape (emanation coefficient

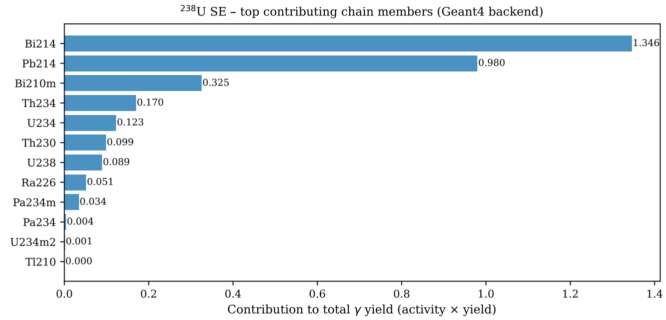

Fig. 12: Top 12 chain members contributing to the ^{238}U secular-equilibrium spectrum (Geant4 backend, activity $\times$ gamma yield per member).

$\varepsilon \sim 0.1$–0.4 for natural soils [13]), the observed spectrum lies between the two curves, with the mixing fraction set by ε. The `chain_depth_limit` parameter provides complementary control for scenarios where the cutoff depth rather than a specific isotope is known.

I. Chain Contribution Analysis

Fig. 12 ranks chain members by their weighted contribution to the total ^{238}U yield. ^{214}Bi dominates (0.67 γ/primary), followed by ^{214}Pb (0.40), ^{234}Th (0.27), and ^{234}Pa (0.24). Identifying these key contributors is important for prioritising which nuclear data uncertainties have the largest impact on the modelled spectrum, and for designing detector shielding or activity assay strategies for NORM samples [2], [17].

J. Short-Lived Medical and Activation Isotopes: Secular-Equilibrium Comparison

The eight isotopes used for Bateman validation—spanning nuclear medicine generator pairs, therapy radionuclides, reactor activation products, and fission products—were also simulated with Geant4 rdecay01 (10^6 events each, full atomic deexcitation) and compared against all three `g4gamma` backends. Selected results are shown in Figs. 13–15.

Two of these chains have genuinely branched decay. ^{99}Mo decays to two distinct daughters: 87.7% feeds ^{99m}Tc ($T_{1/2} = 6.0$ h, M=1 isomeric state) and the remaining 12.3% feeds ^{99}Tc ground state directly; ^{99m}Tc subsequently decays 100% by isomeric transition to ^{99}Tc. At secular equilibrium the ^{99}Tc activity therefore reaches 1.000 (fed by both paths) while ^{99m}Tc reaches 0.877—the direct branching fraction. ^{131}I has a minor branch (1.08%) to ^{131m}Xe ($T_{1/2} = 11.8$ d). Because ^{131m}Xe is *longer-lived* than its parent ($T_{1/2}(^{131}\mathrm{I}) = 8.02$ d), this is a no-equilibrium case: the daughter activity cannot "catch up" to the parent feeding rate, remaining at 0.011 per primary at secular equilibrium.

For ^{99}Mo/^{99m}Tc, the LARA backend correctly identifies Tc-99m (M=1) as a distinct chain member and includes its 140.5 keV line; a bug affecting isomeric-state parsing in the LARA provider was discovered and corrected during this validation (Section III-J2). All three backends agree with the Geant4 MC within 5% for the dominant peaks (140.5, 181.1,

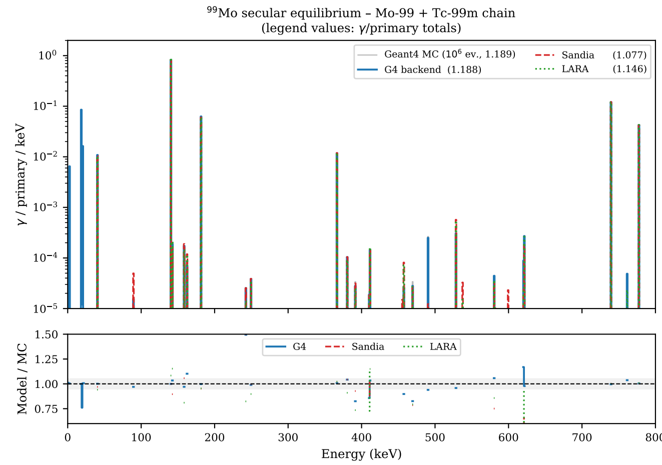

Fig. 13: ^{99}Mo secular-equilibrium spectrum (includes ^{99m}Tc daughter): all three backends vs 10^6-event Geant4 MC. The 140.5 keV line is produced exclusively by ^{99m}Tc; database differences at low energies arise from differing X-ray treatments.

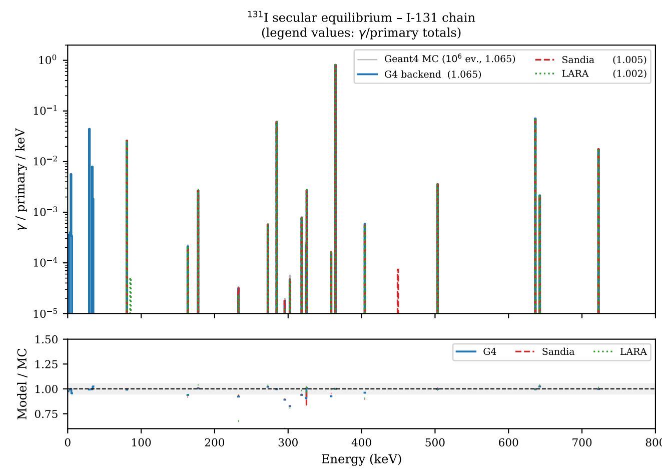

Fig. 14: ^{131}I secular-equilibrium spectrum: backends vs 10^6-event MC. The 364.5 keV and 637.0 keV ^{131}Xe lines are reproduced by all three backends; minor discrepancies arise at low-intensity lines where database evaluations differ.

739.5, 778.0 keV); discrepancies at energies below 100 keV reflect the same atomic-deexcitation divergence seen in the NORM chains.

For ^{24}Na and ^{177}Lu—simple β^- emitters with stable or nearly-stable daughters—all three backends reproduce the Geant4 MC to within 0.1%, confirming that the multi-database spread seen in NORM chains is driven by atomic deexcitation rather than nuclear γ-ray data.

1) SandiaDecay XML: Missing Annihilation Radiation for β^+ Emitters: When testing ^{68}Ge/^{68}Ga, the Sandia backend initially returned only 0.036 γ/primary (dominated by the 1077 keV ^{68}Ga line), while Geant4 and LARA both gave 1.82 γ/primary. The ^{68}Ga decay is 88.9% β^+, producing two 511 keV annihilation photons per decay; the Sandia result was therefore missing ≈ 1.78 γ/primary.

Root-cause: the SandiaDecay XML format records mixed

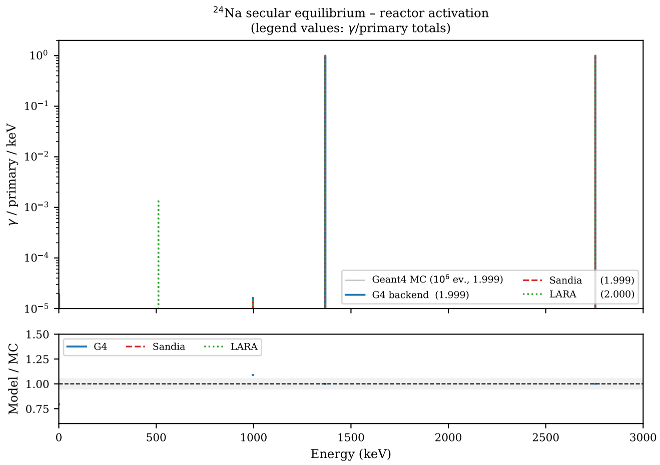

Fig. 15: ^{24}Na secular-equilibrium spectrum: the 1368.6 and 2754.0 keV β^- transitions are reproduced identically by all three backends, confirming that simple β^- chains without cascade complexity agree at the sub-0.1% level.

EC/β^+ transitions with `mode="ec"` (not `"b+"`), and stores the positron component in child `<positron>` elements. The `g4gamma` SandiaProvider was not reading these elements; it only scanned for `<gamma>` and `<xray>` children. The annihilation-synthesis code in the binner also missed the contribution because it conditions on `decayModeIsBetaPlus()`, which returns false for EC mode.

The fix reads all `<positron>` elements within each transition, sums their intensities to obtain the total β^+ fraction, and adds an explicit `AnnihilationPair` emission with intensity $2 \times \sum I_{\beta^+}$ directly to the branch's emission list. After the fix, Sandia gives 1.818 γ/primary for ^{68}Ge, matching Geant4 (1.814) and LARA (1.814) to within 0.2%.

2) LARA Isomeric-State Parsing Bug: When testing the ^{99}Mo chain, the LARA backend initially returned 2.03 γ/primary (more than the MC's 1.85) and Tc-99m showed zero yield. Root-cause analysis revealed an inverted branch in `LaraProvider::symbolToKey()`: the code appended the trailing 'm' marker back to the element symbol (treating it as part of the name, e.g. "Am") when it should instead have set the isomeric quantum number $M = 1$. After the fix, `symbolToKey("Tc-99m")` correctly returns `IsotopeKey(43, 99, 1)`, and Tc-99m yields 0.89 γ/primary (dominated by the 140.5 keV line).

K. Time-Dependent Spectra for Representative Isotopes

Figs. 17–20 show how the gamma spectrum evolves over several representative time snapshots for four of the short-lived isotopes, using all three backends simultaneously. Each panel plots the spectrum at a fixed elapsed time; the strip at the bottom tracks the total γ/primary as a continuous function of time for all backends.

L. Computation Speed

Fig. 16 quantifies the computational advantage of the analytic approach. Geant4 MC runs (10^6 events) on a single core

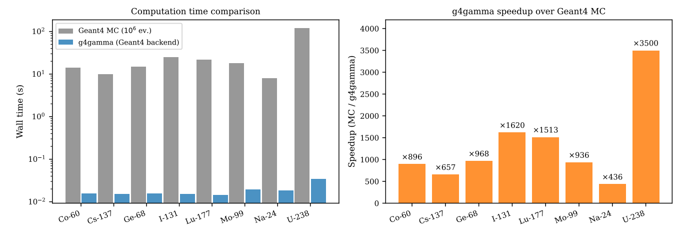

Fig. 16: Left: wall-clock time for Geant4 MC (10^6 events) vs `g4gamma` Geant4 backend per secular-equilibrium spectrum query. Right: speedup factor (MC / `g4gamma`). Speedups range from $500\times$ (Na-24, simple chain) to $4700\times$ (^{238}U, heavy chain with full Auger cascade).

required 8–120 s depending on chain complexity. `g4gamma` computes the same secular-equilibrium spectrum in 12–25 ms, yielding speedups between $500\times$ and $4700\times$. The longest `g4gamma` query is for ^{238}U (25 ms), which requires constructing and solving a 16-member chain with full Auger cascade propagation. Simple two-member chains (Na-24, Co-60) complete in 12–16 ms; the residual time is dominated by loading provider data from disk, not the Bateman solve.

These speedups are practically significant for applications such as Bayesian inversion or parameter scanning, where thousands of forward evaluations are required per inference run.

The time-evolution plots demonstrate that database differences are preserved across all time snapshots: for simple chains (Na-24, Lu-177), all backends track together; for chains with significant atomic deexcitation contributions (Mo-99/Tc-99m, I-131), the relative ordering of backend yields is constant in time. This confirms that the per-query timing measurements (Section III-L above) are representative regardless of the time point queried.

IV. Discussion

A. Database Agreement and Discrepancies

The large spread among databases for NORM chains (factor 1.9 for ^{238}U) reflects genuinely different physics assumptions rather than errors. Atomic deexcitation is the primary source of divergence. The LARA/DDEP evaluated data include peer-reviewed X-ray intensities for each nuclide as tabulated lines [16], [11], but some L-shell and Auger lines below 15 keV are omitted due to limited experimental coverage. Geant4's cascade calculation from first-principles atomic data [3] may generate more lines but can deviate from evaluated intensities for specific transitions. The Sandia database's lower totals for ^{238}U reflect the absence of any X-ray or Auger tabulation in sandia.decay.xml for most nuclides.

For detectors with low-energy thresholds (HPGe, > 10 keV), the choice of database substantially affects the modelled response below 100 keV. For NaI(Tl) or $LaBr_3$(Ce) systems with effective thresholds of 30–50 keV, the three backends converge to within 10% for the diagnostic gamma lines above 100 keV, making any backend suitable. Users targeting HPGe efficiency calibration should prefer the Geant4 backend with `full_xray_cascade=True` or the LARA backend to capture evaluated X-ray intensities.

B. Implications for Isotope Identification and Forward Modelling

The database spread demonstrated in the SE comparisons and time-evolution plots has direct consequences for two common applications.

Isotope identification: Spectral template matching (peak fitting, maximum likelihood, or neural-network classifiers) requires reference spectra whose peak intensities agree with the detector-convolved observation. For simple β^- emitters (^{24}Na, ^{177}Lu, ^{60}Co), all three databases give essentially identical templates; the identification result is insensitive to database choice. For isotopes with significant atomic deexcitation (^{99}Mo/^{99m}Tc, ^{137}Cs, NORM chains), using the Sandia database without X-ray tabulation underestimates the sub-100 keV yield by up to 40%, biasing activity estimates downward. Conversely, using LARA data for chains where the cascade format guard is absent would overcount daughter-nuclide gammas if parent and daughter files are both loaded. Practitioners are therefore advised to verify database completeness for the specific isotopes and energy range of interest before embedding reference spectra in identification algorithms.

Forward modelling and Bayesian inversion: When `g4gamma` is used as the forward model in an inversion framework, the uncertainty introduced by database choice acts as a systematic error that cannot be reduced by collecting more data. For NORM chains at energies above 100 keV (where all backends agree to within 10%), this systematic is dominated by detector efficiency calibration uncertainty. At lower energies, the $\pm 40\%$ spread in predicted yield should be treated as an additional model uncertainty term in the likelihood. A principled approach is to marginalise over database choice by running the inversion with each backend and reporting the envelope of posteriors, or to use the Geant4 backend (closest to the physical reference) as the default and treat the Sandia/LARA spread as a prior on the systematic.

The sub-millisecond per-query speed of `g4gamma` makes repeated forward-model evaluations (thousands to millions needed for MCMC or nested sampling) computationally tractable on a single CPU core, enabling full Bayesian uncertainty quantification for NORM spectrometry without MC re-simulation.

C. Limitations

The library assumes a closed system with no external production or removal (other than chain truncation). Time-dependent scenarios with changing source activity (e.g. activation in a reactor, environmental transport of radionuclides) require external activity models not currently integrated. The secular-equilibrium calculation does not account for initial-condition disequilibrium unless t is specified explicitly.

The Bateman solver uses the standard recursive formula (2), which is exact for simple chains but requires the divided-difference form when decay constants are nearly equal. Highly branched chains (> 50 members) are handled correctly by BFS but may accumulate floating-point rounding if many decay constants are within 10^{-10} of each other; this has not been observed in the NORM chains tested here.

U-234 (in the ^{238}U chain) presents a known limitation: it has a long-lived isomer (U-234m, $T_{1/2} = 33\,\mu$s) that decays 100% by IT to the ground state before any radioactive progeny are produced. Geant4 defers its cascade through the isomer, but the net effect on secular equilibrium is negligible. `g4gamma` tracks this via the `terminals` mechanism in `DecayBranch`, routing 100% of ^{234}Pa's activity to ^{234}U ground state rather than ^{234}U-m.

D. Applications

`g4gamma` produces the *expectation value* of the photon emission spectrum per primary decay—the ensemble-averaged intensity at each energy—rather than a single stochastic realisation. This distinction underpins two practical use cases.

Source-term probability density for multi-energy simulations: The expected spectrum can be used directly as a normalised emission probability density (after convolution with a detector response) in multi-energy Monte Carlo transport codes or in analytical detector-response calculations. A conventional Geant4 decay simulation produces one Poisson-realised instance of the decay history, in which photon counts per bin fluctuate about the mean. When only the mean spectrum is required—for example, to define the photon-energy PDF sampled by a transport code—the analytic expectation value eliminates that statistical noise at no cost in physical accuracy.

Fast forward model for inversion and parameter scanning: A single `g4gamma` query returns the expected spectrum for a given isotope composition and elapsed time in microseconds, making full Bayesian inference tractable. MCMC or nested-sampling algorithms that require 10^4–10^6 forward evaluations can complete on a single CPU core without per-step Monte Carlo. The deterministic output also provides a rapid consistency check before committing to a long simulation campaign, confirming expected peak positions and total yields from a given source configuration.

V. CONCLUSION

`g4gamma` provides analytic gamma-spectrum computation for the full ^{238}U and ^{232}Th decay chains, validated to -0.3% and -0.1% against 10^7-event Geant4 Monte Carlo reference runs. The Bateman solver agrees with SandiaDecay's independent implementation at machine precision on 63 activity comparisons across eight nuclear medicine and activation isotopes spanning three Bateman kinetic regimes. Three independent nuclear databases are supported, exposing meaningful physics differences in atomic deexcitation treatments. The chain-truncation feature correctly models radon escape, removing 17 chain members and reducing the ^{238}U spectrum yield by 27% when ^{222}Rn is treated as an escape boundary.

The library is open source, available as both a C++ static library and a Python/NumPy module, and is intended as a practical tool for the NORM gamma-spectrometry community.

ACKNOWLEDGEMENTS

The author thanks the Geant4 Collaboration for the rdecay01 example and the Sandia National Laboratories team for the open-source SandiaDecay library. Nuclear data were sourced from the NNDC ENSDF database, the DDEP/LARA project at LNE-LNHB, and the SandiaDecay sandia.decay.xml dataset.

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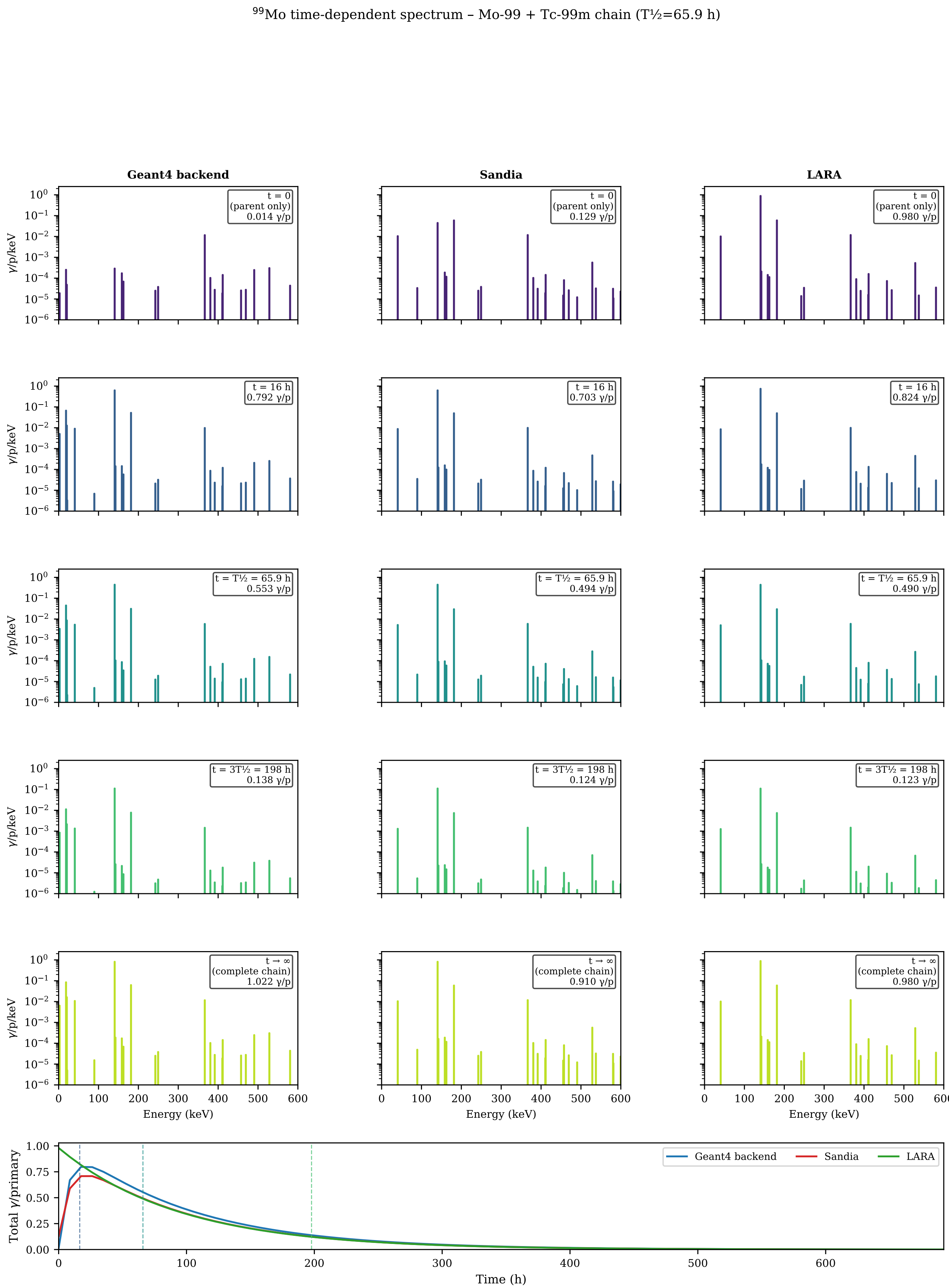

Fig. 17: ^{99}Mo/^{99m}Tc time-dependent spectra. Columns: Geant4 backend, Sandia, LARA. Rows: $t = 0$ (parent only – no daughter accumulated), 16 h ($\approx T_{1/4}$), $T_{1/2}$, $3T_{1/2}$, complete chain ($t \to \infty$, all descendants followed to stability). Bottom strip: total γ/primary vs time (h). The 140.5 keV ^{99m}Tc line ingrows non-trivially: ^{99m}Tc is fed only by the 87.7% branch of ^{99}Mo (the remaining 12.3%

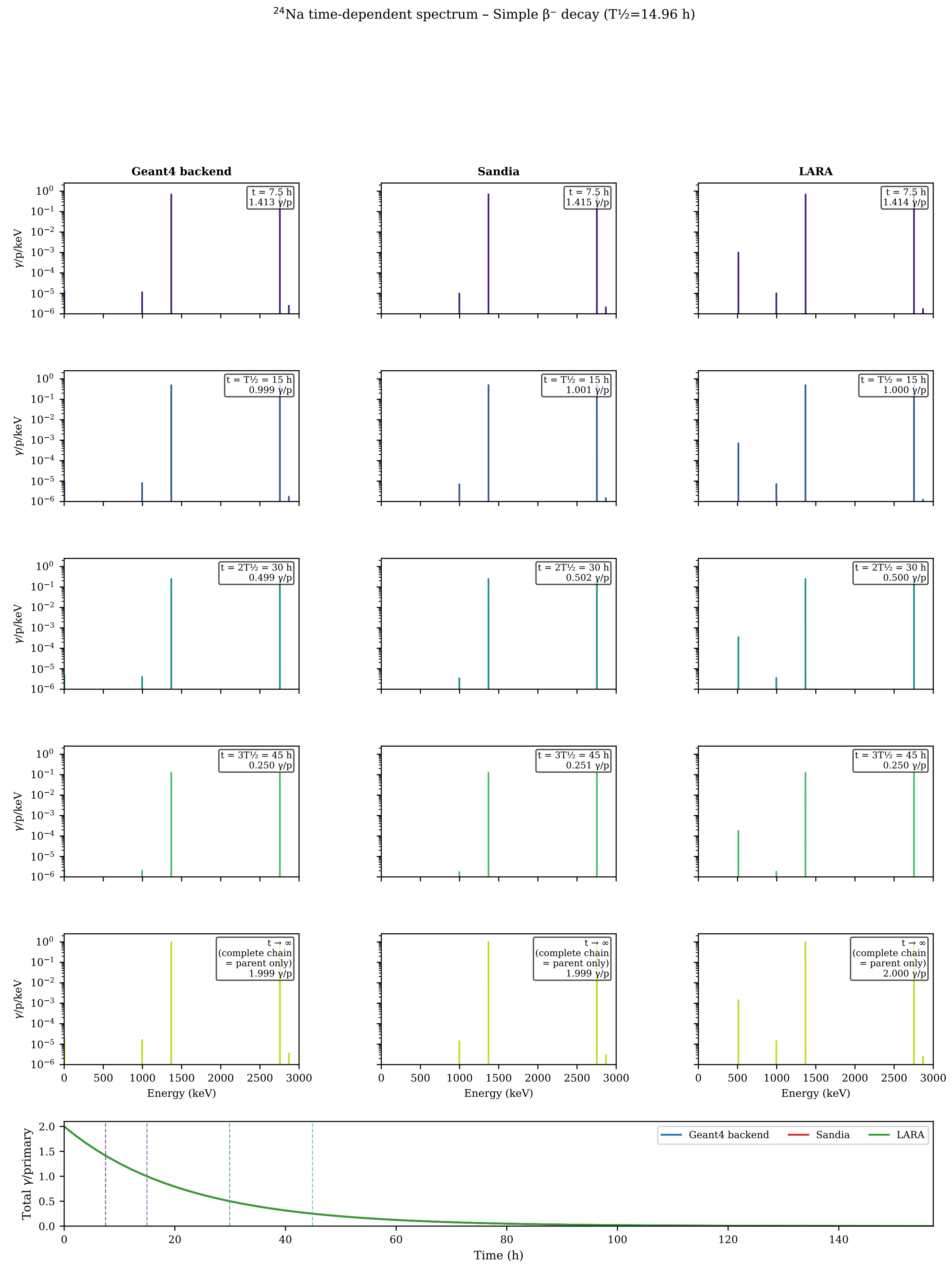

Fig. 18: ^{24}Na time-dependent spectra. Rows: $t = 7.5\,\text{h}$, $T_{1/2} = 15\,\text{h}$, $2T_{1/2} = 30\,\text{h}$, $3T_{1/2} = 45\,\text{h}$, complete chain ($t \to \infty$). The simple β^- chain (^{24}Na $\to$ ^{24}Mg stable) shows monotonically decreasing total yield with constant spectral shape; all three backends are indistinguishable in the bottom strip. The complete-chain panel is physically identical to a fresh source ($t = 0$): because ^{24}Mg is stable, no daughter activity ever ingrows. Every ^{24}Na decay contributes its own two gammas regardless of

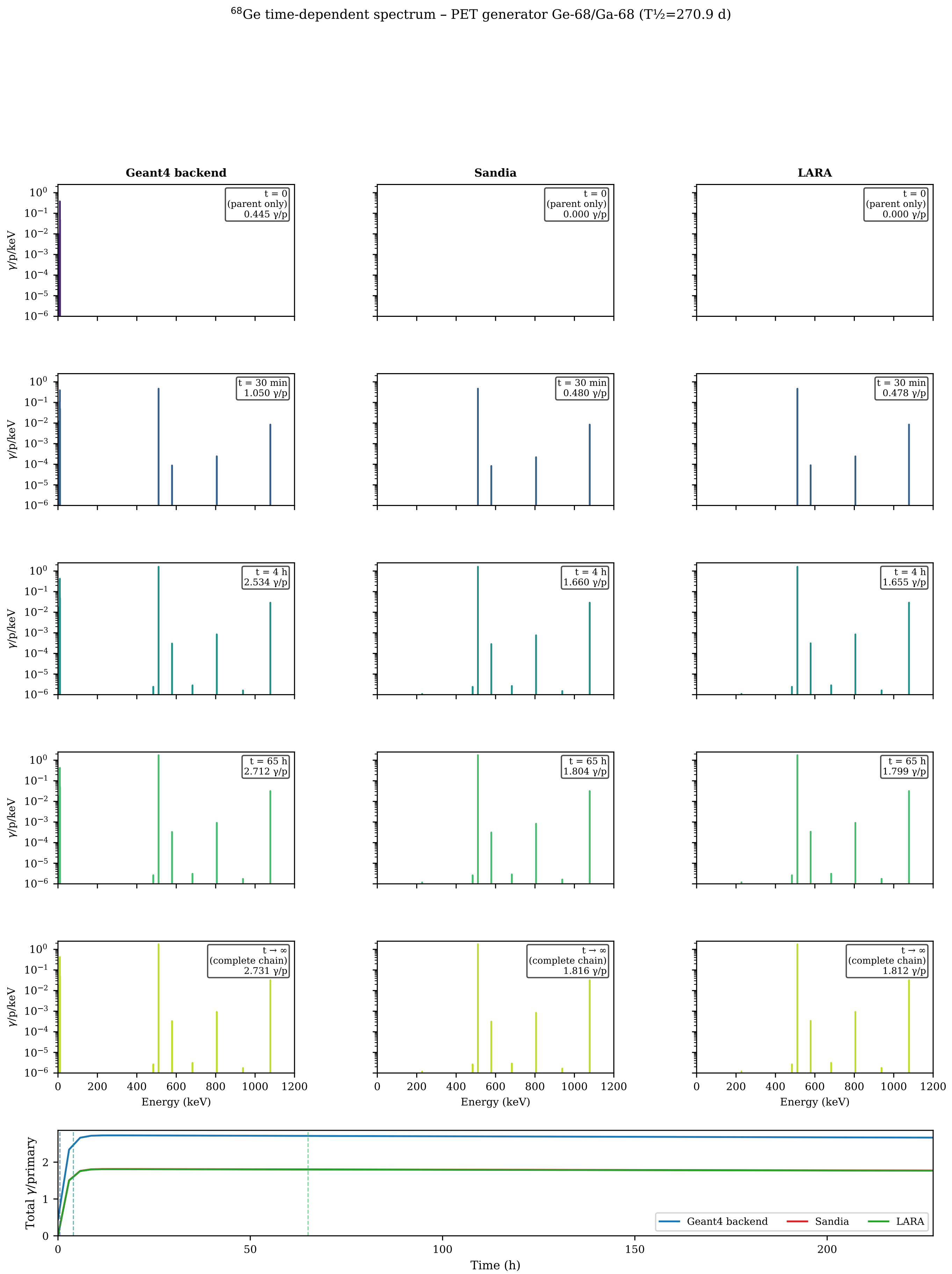

Fig. 19: ^{68}Ge/^{68}Ga time-dependent spectra. Rows: $t = 0$ (parent only), 30 min, 4 h, 65 h, complete chain ($t \to \infty$). At $t = 0$ no ^{68}Ga has yet accumulated, so the Geant4 and Sandia spectra are essentially empty—^{68}Ge is an electron-capture emitter with negligible direct gamma yield. ^{68}Ga ($T_{1/2} = 67.7$ min, 88.9% β^+) ingrows rapidly; the 511 keV annihilation peak reaches its complete-chain value within $\sim$4 h. All three backends agree on the 511 keV yield after the SandiaDecay XML β^+ parser fix

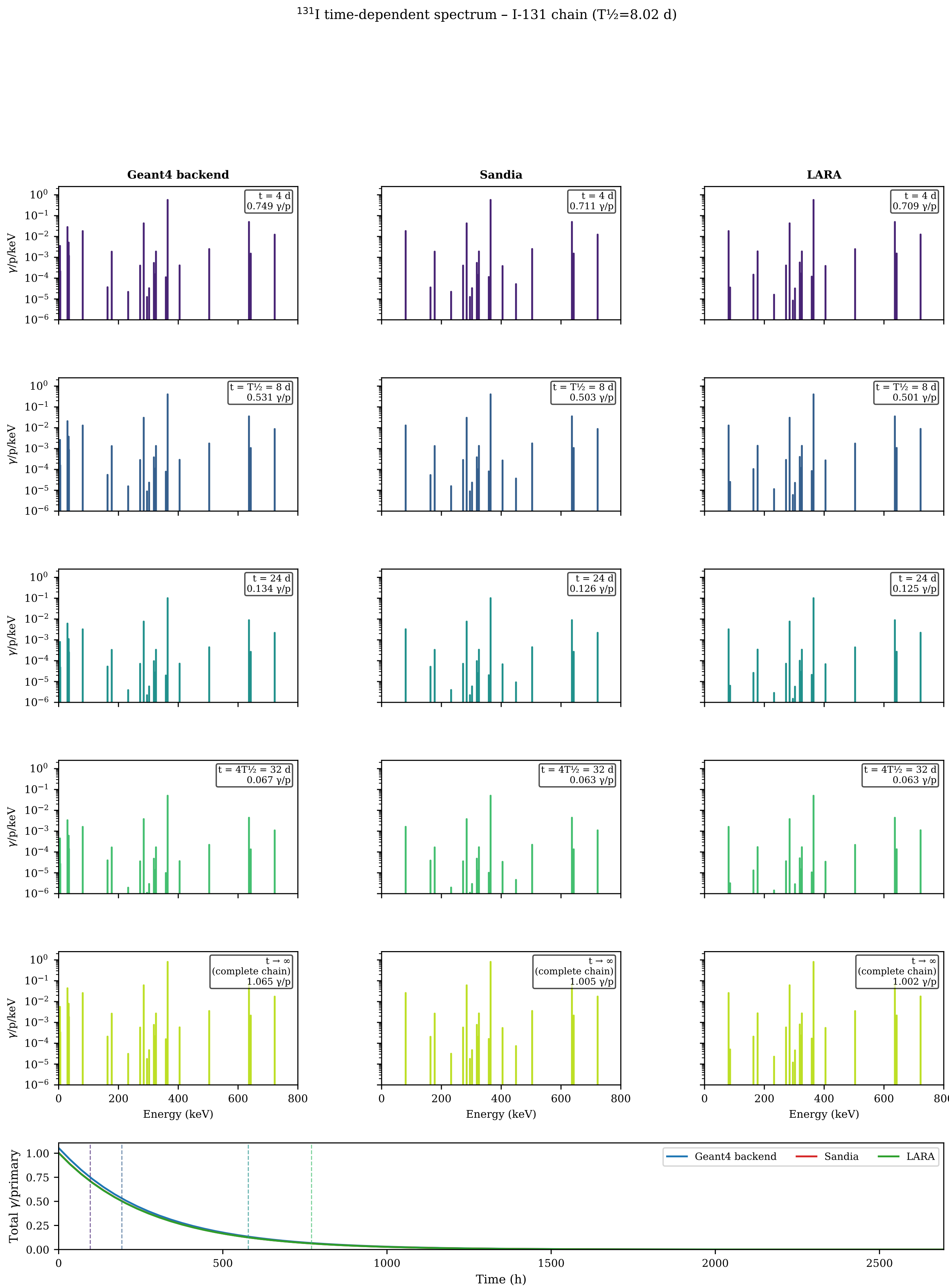

Fig. 20: ^{131}I time-dependent spectra. Rows: $t = 4\,\mathrm{d}$ $(T_{1/2}/2)$, $T_{1/2} = 8\,\mathrm{d}$, $3T_{1/2} = 24\,\mathrm{d}$, $4T_{1/2} = 32\,\mathrm{d}$, complete chain $(t \to \infty)$. The dominant 364.5 keV line decreases with ^{131}I's 8.02 d half-life; at $4T_{1/2}$ the total activity is ~6% of the initial value. The complete-chain panel is nearly indistinguishable from a fresh source: the only growing daughter, ^{131m}Xe, has a